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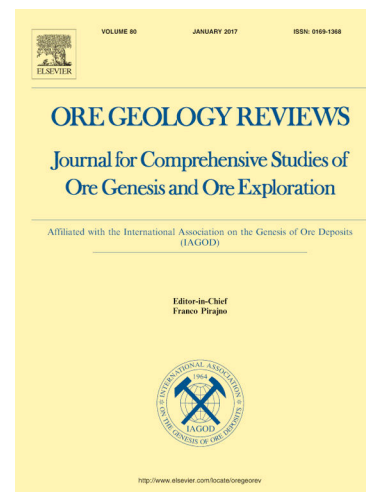
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Sphalerite geochemistry of the Hua'aobaote and Shuangjianshan deposits in the southern Great Xing'an Range, NE China

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Abstract

Sphalerite geochemistry provides new evidence for the mineralization mechanism, ore-formation temperature, genetic type, and prospecting direction of the Hua'aobaote Zn-Pb-Ag deposit and the Shuangjianzishan Ag-Pb-Zn deposit in the southern margin of the Great Xing'an Range. Laser ablation inductively coupled plasma mass spectrometry analyses reveal that most sphalerite grains from both deposits display flat, stable time-resolved signal profiles, indicating that the analyzed domains are largely free of mineral inclusions. Occasional fluctuations in Cu and Ag signals suggest minor inclusions of Cu- and Ag-bearing phases. Sphalerite compositions indicate that divalent cations (e.g., Fe, Mn) are incorporated via direct substitution for Zn^{2+} , whereas cations with other valence states are commonly incorporated through coupled substitution mechanisms. Both deposits contain sphalerite enriched in In and Sn, with average Sn contents of 181 ppm (Hua'aobaote) and 245 ppm (Shuangjianzishan), significantly exceeding those of non-Sn-mineralized Zn-Pb deposits (<10 ppm) but comparable to values from Sn-mineralized systems associated with highly fractionated granites. Such enrichment patterns are indicators of concealed Sn-fertile magmatic-hydrothermal systems. The data therefore suggest that the ore-forming intrusions at both deposits underwent significant differentiation and metal enrichment processes similar to those responsible for Sn mineralization. This supports the view that the currently exploited Ag-Zn-Pb are the distal metallogenic features of larger magmatic-hydrothermal systems that host deeper, high-temperature mineralization (e.g., Sn) closer to the causative intrusion. Application of previous machine learning classification models to multi-element sphalerite data indicates that Hua'aobaote is predominantly of epithermal type, whereas Shuangjianzishan exhibits characteristics of both skarn and epithermal types. Sphalerite geothermometer yields crystallization temperature of 283–336 °C for Hua'aobaote and 270–366 °C for Shuangjianzishan. The obtained temperatures fall below those generally recorded for proximal skarn systems (>400 °C), yet exceed those characteristics of typical MVT deposits (<200 °C). This indicates that Shuangjianzishan represents a transitional magmatic-hydrothermal system with epithermal affinity. Applying this approach to other Ag-Zn-Pb systems with similar sphalerite trace element signatures may help identify concealed Sn orebodies both within the Great Xing'an Range and in surrounding areas.

Keywords: Sphalerite, Ag-Zn-Pb deposits, Exploration implication, Southern Great Xing'an Range

1. Introduction

The southern Great Xing'an Range in NE China is characterized by intense magmatic activity, providing favorable geodynamic and physicochemical conditions for intrusion-related mineralization (Fig. 1). It represents an important province characterized by nonferrous metal mineralization in China, hosting Mo, Cu, Sn, Fe, Zn-Pb, and Ag mineralization (Wu et al., 2011; Zeng et al., 2011; Shu et al., 2016). Ag-Zn-Pb deposits represent an important source of silver and base metals, playing a vital role in modern society (Zhou et al., 2024). The Hua'aobaote and Shuangjianzishan deposits, located in the southern Great Xing'an Range, are typical examples of Ag-Zn-Pb deposits, with numerous studies having focused on their geological settings, metallogenic ages, and tectonic characteristics (Wang et al., 2019; Zhai et al., 2020; Chen et al., 2023; Li et al., 2023a; Wang et al., 2025). However, the ore genesis and formation mechanisms are still weakly understood.

Sphalerite is the dominant sulfide in Zn-Pb mineralization and serves as a significant host for trace

elements (Cook et al., 2009, 2011; Ye et al., 2011). The incorporation of these elements is governed by physicochemical conditions, ore-fluid migration pathways, and fluid-rock interaction, which lead to significant compositional variability and make sphalerite geochemistry a particularly effective proxy for deciphering ore-forming processes (Carrillo-Rosúa et al., 2008; Cook et al., 2009). Recently, sphalerite compositions have been increasingly applied to estimate Zn-Pb mineralization temperature (Keith et al., 2014; Frenzel et al., 2016; Zhao et al., 2024a), trace hydrothermal fluid flow (Cave et al., 2020; Tan et al., 2024), identify the genetic type (Li et al., 2023b; Meng et al., 2024; Zeng et al., 2025), and assess mineralization potential (Gagnevin et al., 2014; Zhang et al., 2022a). Consequently, sphalerite geochemistry has emerged as a key topic in Zn-Pb mineralization research. The Hua'aobaote and Shuangjianzishan deposits host abundant sphalerite, yet detailed geochemical studies of sphalerite from these deposits are lacking.

This study presents a detailed analysis of sphalerite geochemistry from the Hua'aobaote and Shuangjianzishan deposits. We focus on trace element substitution mechanisms in sphalerite, Zn ore formation temperatures, and genetic classification of the deposits. The aim is to utilize sphalerite geochemistry to estimate the mineralization condition and identify the genetic type of these deposits, thereby providing valuable insights to guide exploration.

2. Regional Geological Setting

Prior to the Mesozoic, NE China underwent progressive assembly through subduction-driven accretion and collision of multiple microcontinents within the Paleo-Asian Ocean (Wu et al., 2011). During the Mesozoic, the Paleo-Pacific Plate subduction formed the rudiment of the region (Fig. 1A; Zhou and Wilde, 2013; Wilde, 2015; Shu et al., 2016). Besides the influence of Paleo-Pacific subduction, NE China was affected by post-collisional extension associated with the closure of the Paleo-Asian Ocean. Furthermore, the Paleo-Pacific Plate rollback throughout the Cretaceous further contributed to its tectonic evolution (Ying et al., 2010; Wu et al., 2011; Ouyang et al., 2015; Shu et al., 2016). These processes ultimately resulted in the present-day tectonic framework of NE China, which includes the Erguna massif, Xing'an terrane, Songliao terrane, Liaoyuan terrane, Jiamusi massif, and Nadanhada terrane (Wu et al., 2011; Ouyang et al., 2015). This prolonged and complex tectonomagmatic evolution generated widespread magmatism that is closely linked to significant metal mineralization, a feature widely observed in many metallogenic provinces (Sillitoe, 2010; Meinert et al., 2005; Yang et al., 2016; Wang et al., 2018; Deng et al., 2019; Chang et al., 2019; Yang and Cao, 2024).

Situated on the western wing of NE China, the Great Xing'an Range extends in a near north-northeast direction (Fig. 1A) and is distinguished by voluminous Mesozoic volcanic successions and granitic intrusions (Zhang et al., 2010; Ying et al., 2010; Wu et al., 2011). It spans from south to north across parts of the Xing'an terrane, Songliao terrane, and Erguna massif (Fig. 1A). The Erguna massif is primarily composed of Paleozoic strata, Mesozoic granitoids, and Precambrian granites (Fig. 1A; Shu et al., 2013, 2016). Within the Xing'an terrane, Paleozoic rocks are extensively developed, represented by Early Paleozoic limestones overlain by Late Paleozoic clastic strata (Wu et al., 2011). Magmatism within the Xing'an terrane occurred in two main stages: an early stage represented by Cambrian-Ordovician granites and pegmatites that experienced extensive metamorphism and deformation, and a later stage marked by widely exposed and well-preserved Mesozoic granitic intrusions and volcanic sequences (Zhou and Wilde, 2013). The Songliao terrane

includes the Zhangguangcai Range, the Lesser Xing'an Range, the Songliao Basin, and the southern Great Xing'an Range.

Located in the central region of the Songliao terrane (Fig. 1A), the southern Great Xing'an Range is bounded by several major faults, including the Xilamulun-Changchun, Nenjiang, and Hegenshan-Heihe faults (Fig. 1B). The region comprises over 100 polymetallic deposits enriched in Cu, Mo, Sn, Fe, Zn-Pb, and Ag, forming a major nonferrous metallogenic belt (Shu et al., 2016; Chang et al., 2019; Li et al., 2022; Niu et al., 2022). Structurally, the area is complex, with multiphase folding and faulting exerted strong control on both magmatism and mineralization (Wu et al., 2011). Magmatism is dominated by voluminous Early Cretaceous granitic intrusions and coeval volcanic sequences (Ying et al., 2010; Wu et al., 2011; Shu et al., 2016). The region hosts deposits with a wide range of metal types (Fig. 1B), including Cu (e.g., Lianhuashan, Naoniushan; Yang et al., 2020; Chen et al., 2024), Mo (e.g., Haisugou, Shabutai; Shu et al., 2016; Zhou et al., 2017), Sn-polymetallic (e.g., Weilasituo, Baiyinchagan; Gao et al., 2024; Zheng et al., 2025), Zn-Pb (e.g., Baiyinnuo'er, Haobugao; Shu et al., 2017, 2021; Li et al., 2022; Niu et al., 2022; Zeng et al., 2023), and Ag-polymetallic deposits (e.g., Hua'aobaote, Shuangjianzishan; Zhai et al., 2020; Wei et al., 2024; Wang et al., 2025). Early Cretaceous granitic magmatism played a key role in the genesis of these deposits (Zhang et al., 2010; Ying et al., 2010; Shu et al., 2016).

3. Ore deposit geology

Representative of Ag-polymetallic mineralization, the Hua'aobaote and Shuangjianzishan deposits are situated in the southern Great Xing'an Range (Fig. 1). Together, they contribute significantly to the region's Ag resource endowment and offer important perspectives on regional metallogenic processes (Fig. 2; Fig. 3). Based on field investigations (Fig. 4) and petrographic observations (Fig. 5), the deposit geology and mineral paragenesis were characterized.

The Hua'aobaote Zn-Pb-Ag deposit (45°15'00"–45°16'00"N, 118°56'30"–118°58'00"E) hosts estimated reserves of 0.92 Mt Zn (2.37%), 0.71 Mt Pb (1.99%), and 6,369 t Ag (193 g/t), with minor amounts of 0.06 Mt Cu, 0.01 Mt Sn, and 0.09 Mt Sb (Chen et al., 2023). The wall rocks comprise the Lower Permian Shoushangou Formation, the Upper Jurassic Manketou'ebo Formation and Manitu Formation, and the Neogene Pliocene Baogedawula Formation (Fig. 2; Li et al., 2023a; Wang et al., 2025). The Shoushangou Formation, consisting of siltstone and slate, represents the main host of the ores. The Manketou'ebo Formation consists of tuff and dacite, while the Manitu Formation is mainly composed of andesite and volcanic breccia tuff (Chen et al., 2014). Structural analysis indicates that faults govern the Hua'aobaote deposit, with the NE-trending Meilaote deep fault and associated NW-trending faults forming preferential pathways and spaces for ore emplacement (Fig. 2B, C; Shang et al., 2022).

Magmatic rocks in the mining area mainly comprise granite porphyry and rhyolite porphyry (Fig. 2A, B), while abundant granodiorite porphyry is observed in the drill section (Fig. 2C; Fig. 4A). Zircon U-Pb dating yields two magmatic stages: granodiorite porphyry at 294.8 ± 3.2 Ma and granite porphyry at 132.6 ± 1.4 Ma (Shang et al., 2022), while rhyolite porphyry crystallized at 135.4 ± 0.8 Ma and 134.8 ± 0.8 Ma (Chen et al., 2023). Previous studies conducted cassiterite U-Pb dating of the Hua'aobaote deposit and obtained ages of 135.8 ± 2.7 Ma and 134.9 ± 2.6 Ma (Li et al., 2023a). U-Pb geochronology of cassiterite suggests a temporal overlap with granite porphyry crystallization, indicating that the Hua'aobaote mineralization occurred roughly simultaneously. This supports the

interpretation that granite porphyry acted as the causative intrusion, consistent with regionally extensive magmatism and associated ore formation (Wu et al., 2011; Ouyang et al., 2015; Shu et al., 2016). Ophiolite and ore-hosting serpentinized peridotite are present (Fig. 4B, C). During the final stage of mineralization, clay alteration of malachite and fluorite is observed (Fig. 4D). Orebodies are structurally controlled and occur primarily as parallel veins (Fig. 2B, C). According to their spatial distribution, four ore blocks are distinguished within the deposit: Ag-Zn-Pb mineralization predominates in Block I (south), II (west), and III (northeast), whereas Ag-Cu-Sn mineralization is primarily developed in the central Block IV (Fig. 2B).

The Shuangjianzishan Ag-Zn-Pb deposit contains approximately 18,633 t Ag at a grade of 128.5 g/t, 2.61 Mt Pb at 1.8%, and 0.73 Mt Zn at 0.5%, making it the largest Ag deposit currently known in Asia (Zhai et al., 2020; Wei et al., 2024). The wall rocks comprise the Lower Permian Dashizhai Formation, the Middle Jurassic Xinmin Formation and the Upper Jurassic Manketou'ebo Formation (Fig. 3). The Dashizhai Formation hosts the primary ore-bearing rocks, consisting of andesite, tuff, and silty slate. The Xinmin Formation is primarily composed of dacite and rhyolitic tuff. Pyroclastic rocks and andesite characterize the lithology of the Manketou'ebo Formation. Quaternary rocks are predominantly distributed in valleys throughout most of the area (Fig. 3). In the mining area, faults display NE-, NW-, and nearly NS-trending orientations, and several veins are emplaced along NE-trending faults (Fig. 3A; Wei et al., 2024).

Exposed intrusions in the northwestern mining area are predominantly Yanshanian granite and granodiorite (Fig. 3A; Shi et al., 2024). Minor rhyolitic porphyry and diorite veins are sporadically exposed, primarily within the central and southwestern regions (Fig. 3A, B). At depths between 516 m and 1,080 m, drill hole ZK12-37 intersected an unexposed granite porphyry intrusion (Fig. 4I, J; Zhai et al., 2020), which can be subdivided into a coarse-grained lower facies and a fine-grained upper facies. The fine-grained portion exhibits intense potassic alteration, whereas the coarse-grained facies remain unaltered, dominated by plagioclase and displaying only weak potassic alteration (Zhai et al., 2020). The coarse-grained granite porphyry exhibits minor alteration localized at the contact zone with the fine-grained porphyry, with no alteration observed at depth. In contrast, the fine-grained granite porphyry shows intense potassic alteration and contains quartz-sulfide veins as well as minor magmatic-hydrothermal breccias. Therefore, the fine-grained granite porphyry is inferred to be genetically related to the Ag-Zn-Pb mineralization (Zhai et al., 2020; Shi et al., 2024). The zircon U-Pb age of 134.4 ± 1.0 Ma for the fine-grained granite porphyry, determined by Zhai et al. (2020), is in agreement with Re-Os ages of molybdenite and pyrite (134.9 ± 3.4 Ma and 135.0 ± 0.6 Ma, respectively), indicating that the granite porphyry crystallized contemporaneously with the Ag-Zn-Pb mineralization. The most widespread alteration observed is chloritization (Fig. 4K, L). The orebodies are distributed subparallel to each other along a NE-trending direction (Fig. 3C). The spatial distribution of orebodies allows the deposit to be classified into two ore blocks (Fig. 3B). Orebodies in the Block I are generally smaller in scale. The Block II hosts the principal orebodies, which are primarily stratiform, vein, or lenticular in morphology and occur as a series of NE-trending subparallel zones (Fig. 3C).

Hua'aobaote and Shuangjianzishan are Early Cretaceous Ag-Zn-Pb deposits. The ore minerals include sphalerite, galena, freibergite, chalcopyrite, pyrite, arsenopyrite, and pyrrhotite (Fig. 4E-G, M, N; Fig. 5A-E, J-N). They commonly occur as banded ores or veins (Fig. 4E, N), with chalcopyrite locally exhibiting flame-like textures (Fig. 4G). Some sphalerite grains exhibit zoning, with

individual crystals measuring approximately 200 μm in diameter (Fig. 5A, L). Notably, cassiterite is also present in the Hua'aobaote deposit (Fig. 5F-H). During the post-ore stage, calcite and quartz developed, with quartz vug and platy calcite commonly observed (Fig. 4H, O; Fig. 5I, O).

4. Samples and Methods

For this study, sphalerite samples were obtained from underground workings of the Hua'aobaote and Shuangjianzishan deposits. Trace element determinations of sphalerite were performed using a NWR 193 nm ArF Excimer laser-ablation system coupled to an iCAP RQ (ICPMS) at Guangzhou Tuoyan Analytical Technology Co., Ltd., China. Calibration of the instrument was achieved using the NIST SRM 610 glass reference material. Helium served as the carrier gas (0.7 L/min), supplemented with argon make-up gas at 0.89 L/min. Ablation was performed under operating conditions of 3.5 J/cm² fluence, 6 Hz repetition rate, and 35 μm spot diameter, with each measurement consisting of 40 s of ablation following a 40 s background acquisition. Trace element concentrations were quantified using the multi-external standard total normalization approach, employing NIST SRM610, MASS-1, and YSP as calibration standards (Liu et al., 2008). SRM612 was used as a monitoring standard to evaluate analytical accuracy (Wilson et al., 2002; Danyushevsky et al., 2011). Zn employed as the internal standard, assuming a concentration of 650,000 ppm in sphalerite. Raw isotope data were reduced using the 3D Trace Element data reduction scheme within the IOLITE software (Paton et al., 2011).

5. Results

A total of 84 spot analyses were conducted on sphalerite samples, including 43 from the Hua'aobaote deposit (with 34 valid data) and 41 from the Shuangjianzishan deposit (35 valid data), with the results presented in Table 1 and Table 2. The ranges of trace element concentrations are illustrated in Figure 6. The results show that the types of detectable trace elements in all analyzed samples are consistent, mainly including Mn, Fe, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, and Sb (Table 1; Table 2). The element compositions of sphalerite from the two deposits show certain similarities and differences. In the Hua'aobaote deposit, The content ranges of the 11 trace elements are as follows: Mn (96.7–3619 ppm), Fe (20405–136649 ppm), Co (0.10–45.2 ppm), Cu (9.10–1689 ppm), Ga (0.23–10.7 ppm), Ge (2.20–3.83 ppm), Ag (4.71–295 ppm), Cd (3441–11838 ppm), In (0.10–1528 ppm), Sn (1.75–870 ppm), and Sb (0.20–224 ppm) (Fig. 6; Table 1). A notable feature of this deposit's sphalerite is the relatively high contents of Fe, Cd, and In, with In displaying a particularly large variation. In the Shuangjianzishan deposit, the corresponding trace element content ranges are: Mn (86.7–1429 ppm), Fe (25153–122111 ppm), Co (0.02–201 ppm), Cu (9.84–1524 ppm), Ga (0.19–4.45 ppm), Ge (2.00–3.82 ppm), Ag (1.69–165 ppm), Cd (2928–7392 ppm), In (0.06–749 ppm), Sn (0.20–1411 ppm), and Sb (0.24–31.7 ppm) (Fig. 6; Table 2). In comparison to the Hua'aobaote deposit, the sphalerite here is distinguished by a significantly higher upper limit of Co content, while Cd and In exhibit relatively lower average contents and narrower fluctuation ranges. Additionally, the Sn content is characterized by a broad range, with both a lower minimum value and a higher maximum value than observed in the Hua'aobaote samples.

6. Discussion

6.1 Occurrence of trace elements in sphalerite

During LA-ICP-MS analysis, the presence of large micro-mineral inclusions within the analyzed

domain typically results in time-resolved profiles exhibiting pronounced fluctuations with distinct peaks and troughs. When elements are incorporated into the crystal lattice or homogeneously present as nanoparticles, the corresponding profiles exhibit minimal fluctuations and remain essentially flat (Ye et al., 2011; Zeng et al., 2023). Representative time-resolved profiles of sphalerite from the Hua'aobaote and Shuangjianzishan deposits are shown in Figure 7. Most sphalerite samples from both deposits exhibit flat and smooth signal profiles (Fig. 7A, C). In a few samples, distinct fluctuations or spikes in Cu and Ag signals (Fig. 7B, D) suggest the occurrence of Cu- and Ag-bearing mineral inclusions in sphalerite. To avoid such interference, only sphalerite analyses with flat and smooth profiles were selected. These data reveal compositional trends consistent with isomorphous substitution (Fig. 8).

Because of their comparable ionic radii and oxidation states, Zn^{2+} in sphalerite is commonly substituted by Fe^{2+} , forming ferroan sphalerite (Cook et al., 2009; Belissont et al., 2016; Cave et al., 2020). In both deposits, sphalerite Mn concentrations exhibit a strong positive correlation with Fe (Fig. 8A), indicating that Mn, similar to Fe, substitutes directly for Zn in the sphalerite crystal lattice. Based on these observations, the substitution of divalent cations in sphalerite follows a relatively simple mechanism: $\text{Zn}^{2+} \leftrightarrow (\text{Fe}, \text{Mn})^{2+}$.

Cations such as Cu^+ , Ag^+ , In^{3+} , Sb^{3+} , and Sn^{4+} , which differ from Zn^{2+} in both ionic radius and valence, enter into the sphalerite lattice mainly through coupled substitution in order to maintain charge balance (Cook et al., 2009). For the monovalent cations Ag^+ and Cu^+ , their concentrations display a clear positive correlation (Fig. 8B), consistent with incorporation through the $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{Ag}^+$. Indium is well known to be hosted in sphalerite by a similar substitution mechanism, represented as $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$ (Patrick et al., 1993; Cook et al., 2009; Belissont et al., 2016; Bauer et al., 2019). In the Hua'aobaote and Shuangjianzishan deposits, Cu and In concentrations show an overall positive correlation (Fig. 8C), consistent with this substitution mechanism.

The observed positive correlation between Sb and Ag (Fig. 8D) supports their coupled incorporation into sphalerite via $2\text{Zn}^{2+} \leftrightarrow \text{Ag}^+ + \text{Sb}^{3+}$, as reported in previous studies (Makovicky and Topa, 2018; Liu et al., 2022, 2024). For the tetravalent cation Sn^{4+} , positive correlations are observed between Sn and Ag (Fig. 8E), as well as between Sn and the combined Ag + Cu concentrations (Fig. 8F). These relationships suggest that Sn enters sphalerite via coupled substitution mechanisms such as $3\text{Zn}^{2+} \leftrightarrow 2\text{Ag}^+ + \text{Sn}^{4+}$ or $3\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{Ag}^+ + \text{Sn}^{4+}$. Given that both the Hua'aobaote and Shuangjianzishan deposits are Ag-rich polymetallic systems, the high Ag concentrations in sphalerite likely facilitated Sn incorporation, resulting in its relatively high concentrations.

6.2 Ore-forming temperatures

Previous investigations indicate that temperature plays a critical role in governing trace element incorporation in sphalerite (Möller, 1987; Kelly et al., 2004; Keith et al., 2014; Frenzel et al., 2016). Sphalerite characterized by depleted Fe, In, and Mn but elevated Ga and Ge is indicative of low-temperature Zn mineralization, whereas the opposite signature, with low Ga and Ge combined with enriched Fe, In, and Mn, is commonly associated with high-temperature Zn deposits. Frenzel et al. (2016) applied principal component analysis to sphalerite geochemistry, integrated with fluid inclusion homogenization temperatures, and demonstrated that Fe, Mn, Ga, Ge, and In concentrations are highly sensitive to temperature, thereby establishing the GGIMF as sphalerite geothermometer.

Using this geothermometer, we estimated sphalerite crystallization temperatures of 283–336 °C and 270–366 °C for the Hua'aobaote and Shuangjianzishan deposits, respectively (Fig. 9). These temperature ranges are consistent with those reported from many similar Ag-Pb-Zn polymetallic systems, which typically yield formation temperatures of 250–400 °C (Kelly et al., 2004; Makovicky and Topa, 2018; Liu et al., 2022). Although the estimated temperatures are slightly higher than the fluid inclusion homogenization temperatures previously reported for Hua'aobaote (186–273 °C; Li et al., 2023a) and Shuangjianzishan (210–250 °C; Zhang et al., 2019), they remain broadly consistent with the fluid inclusion results. This minor discrepancy likely reflects analytical uncertainty related to the low Ga and Ge concentrations in sphalerite. Because the sphalerite geothermometer of Frenzel et al. (2016) is based on an inverse correlation between temperature and Ga/Ge contents, slight underestimations of these elements may result in marginally higher estimated temperatures. Overall, the crystallization temperatures suggest that ore formation in both deposits occurred under thermal conditions typical of skarn or epithermal systems, rather than the lower-temperature (<200 °C), sediment-hosted Zn-Pb (MVT) deposits (e.g., Niujiaotang deposit <150 °C, Ye et al., 2012; Nayongzhi deposit <164 °C, Wei et al., 2018).

6.3 Genetic types of mineralization

Sphalerite trace element signatures provide a robust approach for distinguishing among Zn-Pb deposit types (Ye et al., 2011; Li et al., 2023b; Meng et al., 2024; Zeng et al., 2025). Numerous researches have showed that sphalerite formed in different mineralization environments exhibits systematic variations in trace element compositions, reflecting the physicochemical conditions and fluid sources associated with ore formation (Cook et al., 2009; Zeng et al., 2025). The observed compositional variations reflect the fact that trace element incorporation into sphalerite is governed by parameters including temperature, oxygen fugacity, and sulfur fugacity, which differ substantially among deposit types (Kelley et al., 2004; Cook et al., 2009; Keith et al., 2014; Liu et al., 2022).

Cook et al. (2009) first demonstrated that sphalerite from different genetic types of deposits exhibits distinct trace element signatures. Building on this observation, Ye et al. (2011) systematically compared sphalerite compositions across multiple deposit settings and identified clear diagnostic patterns. Distal skarn mineralization commonly contains lower In than proximal zones, providing a useful indicator of mineralization zonation (Zhao et al., 2022; Tan et al., 2024). Sphalerite from syngenetic massive sulfide deposits, which are predominantly hosted in marine volcanic sequences, is enriched in elements such as Ga, In, and Sn. MVT deposits, formed under low temperature basin brine conditions, are generally characterized by higher Ge, Cd, Tl, and As, which preferentially partition into sphalerite in reduced carbonate-hosted settings (Cook et al., 2009). In epithermal deposits, sphalerite typically shows high Ag, Sb, and Tl contents alongside relatively low Co and Mn, reflecting formation from cooler, near-surface hydrothermal fluids differing in physicochemical properties from those in skarn systems (Cooke et al., 2009; Tan et al., 2024; Zeng et al., 2025).

Considering these geochemical characteristics of sphalerite, we applied machine learning algorithms to classify the genetic types of the Hua'aobaote and Shuangjianzishan deposits. We employed two widely-used sphalerite classification models (random forest and support vector machine) developed by Zeng et al. (2025). Zeng et al. (2025) compiled a comprehensive dataset of

4,908 sphalerite analyses from 108 deposits reported in the literature, including concentrations of Mn, Fe, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, Sb, and Pb. The dataset comprises 449 analyses from 14 VMS deposits, 664 from 9 SEDEX deposits, 1,573 from 26 MVT deposits, 746 from 29 skarn deposits, and 1,476 from 30 epithermal deposits. An additional external dataset of 1,518 sphalerite analyses from 23 deposits covering all five deposit types were also incorporated to further expand and validate the model performance. For model construction, Zeng et al. (2025) followed a structured workflow including data pre-processing, dataset splitting, hyperparameter tuning, estimator retraining, model testing, feature importance evaluation, data visualization, and model persistence. Fivefold cross-validation was employed to assess the generalization ability and robustness of the classifiers. Since the accuracy and reliability of these sphalerite classification models have been thoroughly evaluated, we directly applied the same elemental parameters and modeling approach to classify the genetic types of the Hua'aobaote and Shuangjianzishan deposits. Our geochemical data were input into the trained models, which yielded classification results for each dataset and allowed inference of their most probable genetic type of the deposits. Owing to their high accuracy and capacity to resolve complex non-linear relationships among multiple trace elements, these models provide clear advantages over traditional discrimination diagrams or threshold-based approaches (Ye et al., 2011; Li et al., 2023b; Zeng et al., 2025).

For the 34 results from the Hua'aobaote, the random forest model classified 6 as skarn type, 5 as SEDEX type, 4 as MVT type, and 19 as epithermal type (Fig. 10A; Table 1). The support vector machine model classified 7 as skarn type, 1 as SEDEX type, 2 as MVT type, and 24 as epithermal type (Fig. 10B; Table 1). Overall, the results support classifying the Hua'aobaote deposit as mainly epithermal Zn-Pb-Ag, consistent with both its mineralization characteristics and mineral chemistry (Li et al., 2023b; Wang et al., 2025).

For the Shuangjianzishan, a total of 35 sphalerite results were analyzed. The random forest model classified 19 as skarn type, 1 as SEDEX type, 6 as MVT type, and 9 as epithermal type (Fig. 10C; Table 2), whereas the support vector machine model classified 6 as skarn type, 3 as SEDEX type, 24 as epithermal type, and 2 as VMS type (Fig. 10D; Table 2). Collectively, these outcomes indicate that Shuangjianzishan is most consistent with either a skarn-related or an epithermal Zn-Pb-Ag system. Temperature estimates derived from the GGIMFis sphalerite geothermometer (270–366 °C) fall below the values typically associated with skarn mineralization (>400 °C; Meinert et al., 2005; Zhao et al., 2017; Chang et al., 2019), yet are distinctly higher than those of MVT systems (<200 °C; Leach et al., 2010). This suggests that Shuangjianzishan represents a transitional magmatic-hydrothermal system with epithermal affinity. There is no skarn minerals recognized in this deposit, further precluding the possibility of being a skarn deposit.

It is noteworthy that although the machine learning classification results (Fig. 10C) indicate a high probability of Shuangjianzishan being a skarn-type deposit, field observations and petrographic studies reveal no evidence of skarn alteration. This discrepancy likely reflects the fact that the model is based solely on sphalerite elemental composition, which can show significant overlap among different deposit types. Therefore, while the classifier provides useful insights into genetic affinities, its results should be interpreted in conjunction with geological, mineralogical, and alteration information to obtain a more accurate assessment of the deposit type.

6.4 Implications for exploration

Sphalerite is a major carrier of trace elements in Zn-Pb deposits, incorporating Fe, Mn, Co, Cu, Ga, Ge, Ag, In, Sn, Cd, and Sb through complex substitution mechanisms (Huston et al., 1995; Di Benedetto et al., 2005; Cook et al., 2009). Such compositional differences are widely used to distinguish genetic types and assess the fertility of the ore-forming magmatic-hydrothermal systems (Ye et al., 2011; Gagnevin et al., 2014; Zhang et al., 2022a; Chen et al., 2023; Zeng et al., 2025). The Hua'aobaote Zn-Pb-Ag deposit is hosted by Permian clastic rocks and structurally controlled by faults. Within the same region, the Shuangjianzishan Ag-Zn-Pb deposit is hosted in the Lower Permian Dashizhai Formation. Both are classified as epithermal systems, with Ag and Zn-Pb as the primary economic metals.

Granitic intrusions that are highly differentiated commonly exhibit In enrichment, consistent with the elevated In concentrations found in sphalerite of a large number of Sn-bearing skarn deposits (e.g., Murakami and Ishihara, 2013; Gion et al., 2018; Zhao et al., 2022; Kumar et al., 2023). Sphalerite trace element data from the two deposits show significant In enrichment (Fig. 11; Table 1; Table 2), a feature widely regarded as a robust indicator of Sn-fertile magmatic-hydrothermal systems (Gion et al., 2018; Xu et al., 2021; Zhao et al., 2022, 2024b). This geochemical signature provides a key indicator of the concealed metal fertility of the magmatic system beneath the currently exploited Ag and Zn-Pb resources.

It is notable that fluorite occurs in the late-stage alteration assemblage at Hua'aobaote (Fig. 4D). Its presence provides direct mineralogical evidence for F-rich hydrothermal fluids of the magmatic-hydrothermal system. Fluorine is an effective ligand that can complex with a wide range of elements and tends to concentrate in highly differentiated, volatile-rich magmas (Agangi et al., 2010; McPhie et al., 2011). Such F-rich fluids have an enhanced capacity to transport certain metals, including Sn and In, and are commonly associated with Sn-fertile granitoids. Together with the occurrence of cassiterite at Hua'aobaote (Fig. 5A, B, M) and the elevated Sn and In contents in sphalerite (Fig. 6; Fig. 11), the presence of fluorite supports the interpretation that the ore-related magmas underwent extensive fractionation and developed an F-rich volatile phase, thereby enhancing the potential for proximal Sn mineralization (Shu and Deng, 2025). Therefore, the occurrence of fluorite at Hua'aobaote provides extra evidence supporting the Sn fertility indicated by sphalerite geochemistry and further supports the model that distal Ag-Zn-Pb mineralization overlies a deeper, high-temperature Sn-mineralized zone within the magmatic-hydrothermal system.

Sphalerite geochemistry also reveals that Zn-Pb deposits in the Great Xing'an Range, such as Wunu'er, Baiyinnuo'er and Haobugao, contain sphalerite with average Sn contents markedly below 10 ppm, whereas sphalerite from the Sn deposits at Weilasituo and Baiyinchagan exhibits average Sn contents well above 10 ppm (Fig. 11). In comparison, sphalerite from the Hua'aobaote and Shuangjianzishan deposits exhibits significantly higher Sn contents (averaging 181 and 245 ppm, respectively) than those of non-Sn-mineralized Zn-Pb deposits, and these values are comparable to Sn-bearing systems associated with highly fractionated and Sn-fertile granitic intrusions (Fig. 11). The observed geochemical signature indicates that the ore-forming intrusions at Hua'aobaote and Shuangjianzishan deposits likely underwent significant magma differentiation and metal enrichment similar to those of typical Sn deposits, implying that Sn mineralization may occur in closer proximity to the causative intrusions. This is consistent with the fact that there are Sn ores in both Hua'aobaote and Shuangjianzishan (Chen et al., 2023; Wu et al., 2023). The identification of Sn-enriched sphalerite in these Ag-Zn-Pb deposits provides important exploration implications. It

supports a metallogenic model in which distal Ag-Zn-Pb mineralization overlies and surrounds more proximal Sn orebodies closely associated with the causative intrusions. This finding offers potential for identifying high-temperature orebodies (e.g., Sn) in epithermal Ag-Zn-Pb deposits, and may also guide exploration in other similar systems in the Great Xing'an Range and other regions worldwide.

7. Conclusions

Sphalerite in the Hua'aobaote and Shuangjianzishan Ag-Zn-Pb deposits from the southern Great Xing'an Range, NE China incorporates trace elements mainly through simple and coupled substitution mechanisms. Machine learning classification and geological characteristics suggest that both deposits are of epithermal origin. Sphalerite geothermometer yields crystallization temperatures of 283–336 °C for Hua'aobaote and 270–366 °C for Shuangjianzishan, which are consistent with epithermal systems and further support the interpretation that both deposits are of epithermal type. Sphalerite from both deposits is enriched in Sn (181–245 ppm), comparable to values reported from other Sn-fertile granite-related systems in the studied region, supporting the presence of concealed Sn mineralization at depth. Recognition of this geochemical signature serves as a useful indicator in prospecting for proximal, high-temperature Sn-bearing deposits in the southern Great Xing'an Range and other similar metallogenic provinces.

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Figure captions

Figure 1 (A). Simplified geological map showing the tectonic subdivisions of NE China (after Wu et al., 2011). (B). Geological map of structures, magmatic rocks, and mineral deposits in the southern Great Xing'an Range (after Guo et al., 2009 and Li, 2015).

Figure 2 (A). Simplified geological map of the Hua'aobaote area (after Chen et al., 2014). (B). Geological map of the Hua'aobaote Zn-Pb-Ag deposit (after Li et al., 2023a). (C). Geological cross section A–A' of the Hua'aobaote deposit.

Figure 3 Simplified geological map of the Shuangjianzishan area (after Kuang et al., 2014). (B). Geological map of the Shuangjianzishan Ag-Zn-Pb deposit (after Wang et al., 2019). (C). Geological cross section A–A' of the Shuangjianzishan deposit (after Zhai et al., 2020).

Figure 4 Typical hand specimen photographs of the Hua'aobaote and Shuangjianzishan deposits. (A). Banded ores from the main mineralization stage of the Hua'aobaote deposit. (B). Ophiolite from the Hua'aobaote deposit. (C). Quartz vug from the early mineralization stage of the Hua'aobaote deposit. (D). Clay alteration with malachite and fluorite from the late mineralization stage of the Hua'aobaote deposit. (E). Ore-hosting diorite porphyry from the Hua'aobaote deposit. (F). Granite porphyry from the Shuangjianzishan deposit. (G). Disseminated pyritization from the Hua'aobaote deposit. (H). Ore-hosting serpentinitized peridotite from Hua'aobaote deposit. (I). Chalcopyrite ore with flame-like texture from the Hua'aobaote deposit. (J). Mudstone with chloritization from the Shuangjianzishan deposit. (K). Ore-hosting granodiorite porphyry from the Shuangjianzishan deposit. (L). Massive sulfide ore comprising chalcopyrite and sphalerite from the Shuangjianzishan deposit. (M). Platy calcite from the Shuangjianzishan deposit. (N). Sulfide-quartz vein from the main mineralization stage of the Shuangjianzishan deposit. (O). Volcanic breccia with chloritization from the Shuangjianzishan deposit. Abbreviations: Bi = biotite, Cal = calcite, Ccp = chalcopyrite, Chl = chlorite, Fl = fluorite, Gn = galena, Kfs = K-feldspar, Kln = kaolinite, Mlc = malachite, Pl = plagioclase, Py = pyrite, Qtz = quartz, Sp = sphalerite, Srp = serpentine.

Figure 5 Typical photomicrographs of the Hua'aobaote and Shuangjianzishan deposits. (A). Cassiterite and arsenopyrite from the early mineralization stage of the Hua'aobaote deposit. (B). Cassiterite from the early mineralization stage of the Hua'aobaote deposit. (C). Arsenopyrite from the Shuangjianzishan deposit. (D). Paragenetic association of chalcopyrite, pyrite and pyrrhotite from the Shuangjianzishan deposit. (E). Zoned sphalerite from the Shuangjianzishan deposit. (F). Galena and sphalerite with chalcopyrite exsolution from the Shuangjianzishan deposit. (G). Banded sphalerite from the Hua'aobaote deposit. (H). Galena and stibnite from the Hua'aobaote deposit. (I). Paragenetic association of sphalerite, chalcopyrite and freibergite, with late-stage pyrite from the Hua'aobaote deposit. (J). Paragenetic association of chalcopyrite, freibergite and argentite from the Shuangjianzishan deposit. (K). Sphalerite and late-stage pyrite from the Hua'aobaote deposit. (L). Pyrite, sphalerite and freibergite from the Hua'aobaote deposit (BSD). (M). Pyrite and cassiterite from the Hua'aobaote deposit (BSD). (N). Cathodoluminescence (CL) image of zoned quartz with calcite from the Shuangjianzishan deposit. (O). Cathodoluminescence (CL) image of zoned quartz from the Hua'aobaote deposit. Abbreviations: Apy = arsenopyrite, Arg = argentite, Cal = calcite, Ccp = chalcopyrite, Cst = cassiterite, Frb = freibergite, Gn = galena, Po = pyrrhotite, Py = pyrite, Qtz = quartz, Snt = stibnite, Sp = sphalerite.

Figure 6 Box-and-whisker plots of the trace element compositions of sphalerite from the Hua'aobaote and Shuangjianzishan deposits.

Figure 7 Representative time-resolved LA-ICP-MS depth profiles of sphalerite from the Hua'aobaote and Shuangjianzishan deposits. (A, B) Hua'aobaote deposit: smooth signal in (A) and irregular signals in (B). (C, D) Shuangjianzishan deposit: smooth signal in (C) and irregular signals in (D).

Figure 8 Binary plots showing inter-element correlations and substitution mechanisms in sphalerite. (A). Fe vs. Mn, (B). Cu vs. Ag, (C). Cu vs. In, (D). Ag vs. Sb, (E). Sn vs. Ag, and (F). Sn vs. (Cu + Ag).

Figure 9 Frequency distribution plots of the estimated formation temperatures of sphalerite from the Hua'aobaote (A) and Shuangjianzishan (B) deposits.

Figure 10 Machine learning model's classification pie chart of the Hua'aobaote (A, B) and Shuangjianzishan (C, D) deposits. The numerical values in the pie chart represent the number of samples classified by the machine learning models as belonging to a corresponding type. Abbreviations: RF = random forest, SVM = support vector machine.

Figure 11 Box-and-whisker plots showing the sphalerite Sn contents from the Hua'aobaote and Shuangjianzishan deposits. Also shown for comparison are samples from the Wunuer Zn-Pb-Ag deposit (data from Fan et al., 2020), the Baiyinnuo'er Zn-Pb deposit (data from Dai, 2022), the Haobugao Zn-Pb deposit (data from Dai, 2022), the Weilasituo Sn-polymetallic deposit (data from Gao et al., 2024), and Baiyinchagan Sn-Zn-Pb-Ag deposit (data from Zhang et al., 2022b). Note that the Hua'aobaote and Shuangjianzishan deposits are comparable to the Sn-bearing Weilasituo and Baiyinchagan deposits, and significantly higher than the non-Sn-bearing Wunuer, Baiyinnuo'er, and Haobugao deposits, suggesting potential for Sn mineralization in the Hua'aobaote and Shuangjianzishan deposits.

Table captions

Table 1 Trace element compositions, estimated formation temperatures of sphalerite, and machine learning classification results from the Hua'aobaote deposit

Table 2 Trace element compositions, estimated formation temperatures of sphalerite, and machine learning classification results from the Shuangjianzishan deposit

Table1 Trace element compositions, estimated formation temperatures of sphalerite, and machine learning classification results from the Hua'aobaote deposit

Sample No.	Fe	Zn	Mn	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	T	RF	SVM
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	wt %	wt %	pp m	pp m	pp m	pp m	pp m	pp m	pp m	pp m	pp m	pp m	° C		
HT-2-1-5	5.6 8	62. 7	14 1	2.6 0	52 9	0.6 7	3.3 0	27. 5	873 1	45. 2	33. 0	2.2 1	31 0	skarn	skarn
HT-2-1-6	10. 9	57. 1	33 3	B DL	45 2	0.2 5	3.2 4	64. 3	863 5	B DL	B DL	25. 4	29 0	SEDE X	epithermal
HT-2-1-7	12. 5	55. 3	29 1	0.3 3	97 8	0.8 4	3.2 3	11 6	412 2	0.1 0	65 0	88. 9	29 5	skarn	epithermal
HT-2-1-8	11. 6	56. 4	50 1	0.1 0	30 9	0.6 1	3.8 3	34. 6	456 4	B DL	B DL	22. 1	28 3	epithermal	epithermal
HT-2-1-10	11. 1	56. 6	61 8	0.0 4	15 05	0.5 4	3.3 2	29 5	456 1	B DL	B DL	22 4	28 8	SEDE X	epithermal
HT-2-1-11	8.9 1	59. 5	41 7	0.1 2	50 8	0.2 3	3.4 3	38. 3	442 4	B DL	B DL	13. 8	28 9	epithermal	epithermal
HT-2-1-12	6.5 5	61. 7	23 5	1.1 5	62 4	0.7 1	3.0 3	23 7	833 4	16 2	43 6	10 2	32 7	epithermal	epithermal
HT-64-1	8.1 9	60. 4	86 4	0.0 5	46. 8	0.9 8	3.5 2	21. 7	937 7	0.2 5	14. 2	1.3 1	30 1	epithermal	skarn
HT-64-2	5.9 7	62. 9	36 3	1.0 8	78. 3	0.6 7	3.2 4	34. 3	954 0	52. 5	87 0	2.2 5	32 3	epithermal	epithermal
HT-64-3	4.7 2	64. 0	24 4	0.5 9	13 0	0.8 9	2.5 8	15. 8	101 24	24. 1	9.1 2	0.9 1	30 8	epithermal	epithermal
HT-64-4	6.2 2	62. 6	53 2	2.2 5	89 4	0.4 3	3.5 4	43. 1	103 16	26. 5	B DL	5.1 9	32 8	epithermal	SEDE X

HT-64-5	4.3 9	64. 2	30 3	1.4 0	11 21	0.2 5	3.0 3	17 7	115 56	25 9	30 8	11 1	33 7	epith mal	epith mal
HT-64-6	4.8 4	63. 8	31 4	3.8 2	20 9	0.5 0	3.6 8	16. 0	118 38	98. 9	10. 9	0.2 0	32 2	epith mal	skarn
HT-64-7	4.3 5	64. 3	25 6	5.9 1	19 7	1.0 5	3.7 7	12. 3	928 6	22 7	B DL	0.2 4	31 4	skarn	skarn
1206-43-1	10. 2	55. 3	23 16	45. 2	18 0	1.3 2	B DL	22. 4	521 9	17. 7	96. 6	B DL	—	epith mal	epith mal
1206-43-2	10. 6	55. 1	27 82	27. 4	11 3	1.2 1	2.2 5	28. 1	523 7	2.2 5	20. 6	1.5 9	33 5	epith mal	epith mal
1206-43-3	9.9 7	56. 1	26 84	42. 8	33. 6	1.3 0	B DL	9.6 1	542 1	9.9 1	19. 8	B DL	—	skarn	epith mal
1206-43-4	10. 6	55. 2	32 60	30. 5	15 7	1.7 1	2.2 5	21. 8	559 8	4.3 2	93. 3	B DL	33 6	epith mal	epith mal
1206-43-5	9.7 0	56. 3	23 96	42. 2	48. 9	0.7 5	B DL	8.4 9	509 2	10. 9	35. 1	B DL	—	skarn	epith mal
1206-43-6	11. 2	54. 5	20 14	24. 5	24 2	2.3 4	2.3 4	22. 2	538 3	3.9 3	76. 4	B DL	32 7	SEDE X	epith mal
1206-30-1	13. 1	52. 8	31 63	3.8 8	18 9	8.7 0	2.2 5	28. 1	447 3	0.3 0	2.4 5	5.0 5	30 5	epith mal	epith mal
1206-30-2	12. 3	53. 6	58 9	9.2 4	9.1	3.5 1	3.4 7	5.0 4	496 7	0.3 7	2.1 4	B DL	29 2	epith mal	epith mal
1206-30-3	12. 5	53. 2	27 95	14. 6	26 1	4.8 2	2.3 4	24. 6	492 4	23. 1	62. 2	2.2 4	33 5	SEDE X	epith mal

1206-30-5	13.2	52.5	33.97	2.56	21.7	9.46	2.33	16.4	4483	0.43	10.8	1.40	306	epithermal	epithermal
1206-30-6	13.7	51.9	36.19	3.17	25.8	10.7	2.34	16.6	5021	0.44	1.75	1.36	306	SEDEX	epithermal
2302-1-1	2.04	64.7	13.1	B DL	78.6	0.84	3.00	14.8	3489	15.28	8.01	B DL	308	MVT	epithermal
2302-1-2	10.4	55.6	34.12	B DL	61.5	2.03	2.20	5.69	4276	2.54	33.5	1.24	331	skarn	skarn
2302-1-3	4.83	61.8	15.47	B DL	16.89	2.61	2.21	30.6	3657	32.8	18.3	B DL	320	epithermal	epithermal
2302-1-4	2.56	64.2	68.1	0.15	45.8	1.72	3.70	12.8	3932	17.0	10.5	0.99	293	epithermal	skarn
2302-1-5	3.86	63.0	10.56	B DL	11.9	1.23	3.16	8.43	3829	15.0	27.0	B DL	325	epithermal	epithermal
2302-1-7	2.72	64.3	96.7	B DL	12.1	1.16	2.73	4.71	3441	20.4	5.61	0.59	296	MVT	MVT
2302-1-10	2.50	63.8	10.4	B DL	11.09	2.32	3.00	21.1	4068	13.57	16.1	5.72	297	epithermal	epithermal
2302-1-11	3.03	63.7	15.3	0.44	34.3	1.36	3.02	18.7	3611	16.7	9.09	3.19	299	MVT	skarn
2302-1-12	2.52	64.3	10.1	B DL	38.9	1.73	3.00	18.2	3645	70.4	9.39	3.63	296	MVT	MVT

Abbreviations: BDL = Below the detection limit, RF = random forest, SVM = support vector machine.

–: Due to Ge contents being below the detection limit, the formation temperature of sphalerite could not be determined.

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Table 2 Trace element compositions, estimated formation temperatures of sphalerite, and machine learning classification results from the Shuangjianzishan deposit

Sample No.	Fe wt %	Zn wt %	Mn pp m	Co pp m	Cu pp m	Ga pp m	Ge pp m	Ag pp m	Cd pp m	In pp m	Sn pp m	Sb pp m	T ° C	RF	SVM
0605-45-1	2.5 2	63. 9	88. 7	14. 7	10. 7	0.4 6	2.8 4	1.6 9	44 96	2.1 7	7.4 7	B DL	27 7	MVT	epithermal
0605-45-2	2.5 2	64. 0	86. 7	13. 7	13. 7	0.3 4	2.8 4	2.3 0	46 30	2.5 7	9.5 7	B DL	28 1	MVT	epithermal
0605-45-3	2.5 8	63. 6	97. 2	11. 6	13 25	1.0 1	3.1 2	71. 4	41 77	10 2	11 06	16. 8	29 1	epithermal	epithermal
0605-45-4	3.0 8	63. 6	15 1	22. 3	35. 1	0.7 9	2.8 2	4.2 2	37 55	38. 9	10. 4	0.7 4	29 7	SEDEX	skarn
0605-45-5	4.8 5	61. 5	26 4	56. 6	39 5	3.0 9	3.6 9	13. 6	39 13	87. 6	30 2	0.3 9	29 8	epithermal	epithermal
0605-45-6	4.6 3	61. 8	29 6	84. 1	29 8	1.2 2	3.5 9	17. 2	36 88	38 9	70. 9	1.1 3	31 9	skarn	skarn
0605-45-7	4.7 0	62. 0	28 3	48. 5	40 6	0.7 6	3.5 8	15. 6	36 60	74 9	18. 5	B DL	32 8	epithermal	epithermal
0605-45-8	5.7 4	60. 5	27 9	36. 6	75 0	4.3 5	3.6 6	31. 3	33 90	6.9 8	67 8	B DL	28 3	epithermal	epithermal
0605-45-9	5.6 2	60. 7	26 4	34. 8	55 5	3.9 3	3.6 8	39. 4	32 81	6.6 3	45 8	0.9 4	28 3	skarn	epithermal

0605-45-10	5.6 7	60. 6	27 5	28. 0	91 9	3.3 9	3.6 8	47. 9	34 31	6.2 9	81 0	3.1 6	28 5	skarn	epithermal
0605-45-11	5.7 5	60. 4	29 3	30. 7	15 24	3.8 6	3.6 5	63. 3	34 44	9.2 9	14 11	0.4 5	28 7	skarn	epithermal
0605-45-12	5.6 9	60. 7	30 1	28. 7	13 67	3.5 1	3.6 5	75. 2	35 44	6.1 6	81 7	7.6 7	28 5	skarn	epithermal
0605-45-13	5.7 0	60. 7	29 0	34. 5	11 06	4.0 7	3.6 4	15 6	35 11	8.0 0	81 3	1.3 0	28 5	skarn	epithermal
0605-45-14	5.4 1	61. 2	24 0	29. 2	24 9	4.4 5	3.2 1	14. 8	33 00	7.0 8	22 4	B DL	28 1	skarn	epithermal
0605-45-15	5.5 5	60. 9	24 7	29. 5	39 1	3.4 9	3.2 2	22. 9	33 54	7.6 7	35 2	B DL	28 6	skarn	epithermal
0605-45-16	5.7 0	60. 6	27 4	33. 4	86 7	4.3 1	3.6 8	32. 4	34 40	9.4 1	79 8	B DL	28 4	epithermal	epithermal
0605-16-1	4.8 7	61. 7	26 3	B DL	9.8 4	0.3 2	3.0 8	4.1 3	32 60	0.0 6	8.7 7	4.7 0	28 4	MVT	MVT
0605-16-2	4.3 7	62. 1	14 5	B DL	38. 6	0.4 0	2.7 3	5.7 1	42 05	0.1 6	34. 3	2.6 4	28 0	MVT	MVT
0605-16-3	5.3 2	61. 3	17 2	0.0 2	11 5	1.8 9	3.2 9	25. 4	38 83	0.4 5	10 6	15. 9	27 1	MVT	epithermal
0605-16-6	3.9 2	62. 7	20 4	B DL	34 5	0.6 1	3.0 0	25. 6	31 35	1.3 8	32 1	9.7 0	28 8	skarn	epithermal
0605-16-11	12. 2	53. 8	65 1	1.1 6	34 8	1.1 7	3.8 2	13. 6	29 28	1.3 7	31 8	B DL	31 3	skarn	skarn

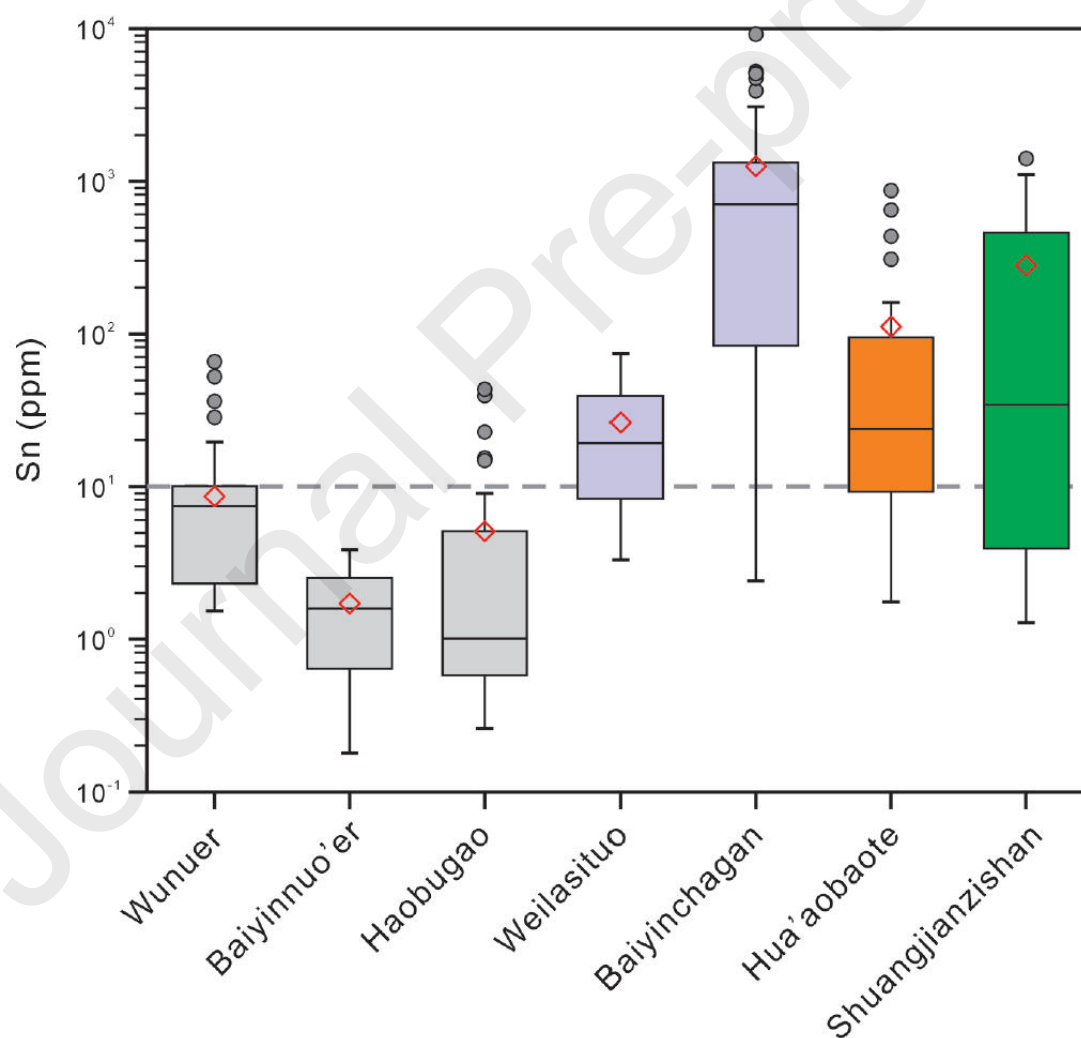
0605-16-12	4.8 4	61. 4	20 6	B DL	15. 7	0.3 5	3.0 9	3.4 5	35 95	0.1 0	14. 5	1.9 3	28 3	MVT	MVT
SJ-200-1	9.4 8	56. 8	14 29	14 5	43. 4	0.2 5	2.0 0	6.5 1	73 08	34. 3	1.2 9	B DL	36 2	skarn	epithermal
SJ-200-2	8.5 4	57. 3	13 25	20 1	33. 5	0.3 3	3.2 3	7.8 7	71 87	77. 0	1.6 6	B DL	35 5	skarn	epithermal
SJ-200-3	9.2 6	56. 7	13 84	15 5	31. 8	B DL	B DL	6.2 6	73 92	32. 7	1.6 9	B DL	—	skarn	epithermal
SJ-200-4	8.7 9	57. 6	13 66	15 7	34. 6	0.2 8	2.0 0	7.6 9	72 99	56. 6	1.6 6	B DL	36 1	epithermal	epithermal
SJ-200-5	6.1 2	60. 6	35 3	65. 3	20. 5	0.4 9	3.2 9	3.8 1	60 94	54. 4	1.5 9	B DL	32 6	epithermal	epithermal
SJ-200-6	3.2 9	63. 4	24 5	60. 2	37. 8	B DL	2.9 1	16 5	58 29	59. 8	22. 1	31. 7	36 7	epithermal	epithermal
SJ-200-7	3.9 3	63. 0	41 5	91. 2	36. 3	B DL	B DL	4.6 2	61 88	11 2	B DL	B DL	—	skarn	epithermal
SJ-200-8	4.6 6	61. 7	54 1	43. 0	17. 6	B DL	B DL	4.1 7	56 74	51. 6	B DL	B DL	—	skarn	skarn
SJ-200-9	4.5 5	62. 0	61 8	88. 7	23. 5	B DL	B DL	4.2 8	61 92	72. 9	B DL	B DL	—	skarn	skarn
SJ-200-10	4.3 9	61. 9	88 7	59. 0	29. 0	0.1 9	3.1 7	4.6 7	64 27	87. 0	B DL	B DL	34 4	skarn	VMS
SJ-200-11	7.7 0	58. 2	10 34	84. 7	65. 9	0.6 4	3.4 1	8.1 5	69 28	18 1	3.9 4	0.3 5	34 6	skarn	VMS

12	0	7	40	6	0	8	2	3	06	0	4	DL	0	mal	mal
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SJ-200-	9.2	56.	13	14	32.	0.2	2.0	4.6	73	34.	1.4	0.2	36	skarn	skarn
13	0	8	70	5	6	7	0	3	72	0	7	4	0		

Abbreviations: BDL = Below the detection limit, RF = random forest, SVM = support vector machine.

–: Due to Ge contents being below the detection limit, the formation temperature of sphalerite could not be determined.



Highlights

- Hua'aobaote and Shuangjianzishan are typical epithermal deposits
- Sphalerite crystallized at 283–336 °C in Hua'aobaote and at 270–366 °C in

Shuangjianzishan

- High Sn contents in sphalerite indicate presence of concealed Sn mineralization at depth
- Sphalerite geochemistry provides evidence for mineralization temperature, type, and mechanism, and prospecting direction

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.